Historical SN and their properties

TABLE 11.1 Supernovas Observed in the Galaxy by the Naked Eye

Date (A.D.)	*Constellation*	*Apparent Brightness*	*Distance* (*kpc*)	*Observers*
185	Centaurus	Brighter than Venus (−6)	2.5	Chinese
369	Cassiopeia	Brighter than Mars or Jupiter (−3)	10	Chinese
1006	Lupus	Brighter than Venus (~~−5~~) -10	3.3	Chinese, Japanese, Korean, European, and Arabian
1054	Taurus (Crab Nebula)	Brighter than Venus (−5)	2	Chinese, Southwestern Indian, and Arabian
1572	Cassiopeia	Nearly as bright as Venus (−4)	5	Tycho and many others
1604	Ophiuchus	Between Sirius and Jupiter (−2)	6	Kepler, Galileo, and many others

Source: Adapted from a table compiled by W. C. Straka.

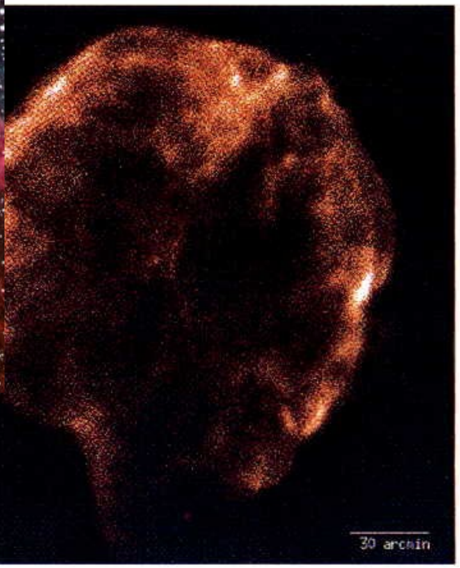
Total energy released ~10^{54} erg in a few hours

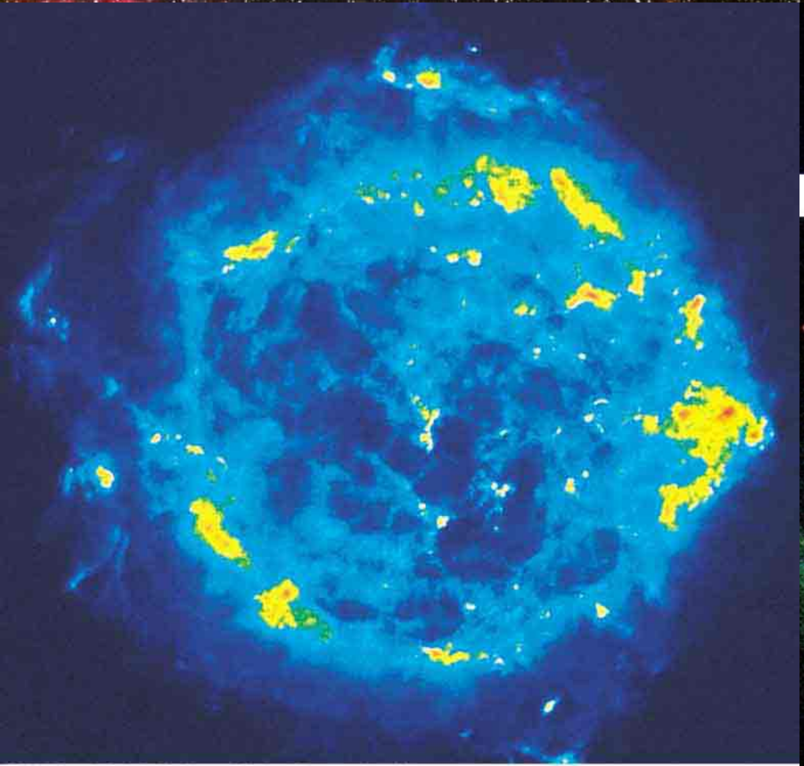
TABLE 11.2 Properties of Supernovas

	Type I	*Type II*
Ejected mass (solar masses)	0.5	5
Velocity of ejected mass (km/sec)	10,000	5000
Total kinetic energy (J)	5×10^{43}	10^{44}
Visual radiated energy (J)	4×10^{42}	10^{42}
Maximum absolute magnitude	−19 to −20	−17
Frequency	1 in 60 yr	1 in 40 yr

Source: R. A. Chevalier, "The Interaction of Supernovae with the Interstellar Medium," *Annual Reviews of Astronomy and Astrophysics*, vol. 15 (1977).

Supernova Remnants





Neutron stars

Stellar Graveyards

Normal stars

These stars do not collapse due to ***self-gravity*** because of the ***thermal pressure*** from the heat generated by ***thermonuclear reactions*** at their centers. When their ***nuclear fuel*** is ***exhausted*** they change form.

Stellar corpses

Less massive stars like the ***Sun*** become ***white dwarfs***, about the size of the Earth.

More massive stars undergo catastrophic collapse leading to ***supernova explosions*** and the expulsion of most of their mass. Their ***cores*** may collapse into ***neutron stars*** or ***black holes***, or perhaps are also disrupted.

Neutron Star Exotica

Neutron Star Gravity

A neutron star's ***mass*** equals the Sun's, but its ***diameter*** is only 20 kilometers or 12.5 miles, so a neutron star's ***gravitational field*** is immensely strong. Dropping an object from space onto a neutron star releases ***~212 million*** times as much energy as dropping the same object from space onto the Earth. In fact dropping matter from space onto a neutron star is releases more energy than ***nuclear fusion*** of that matter.

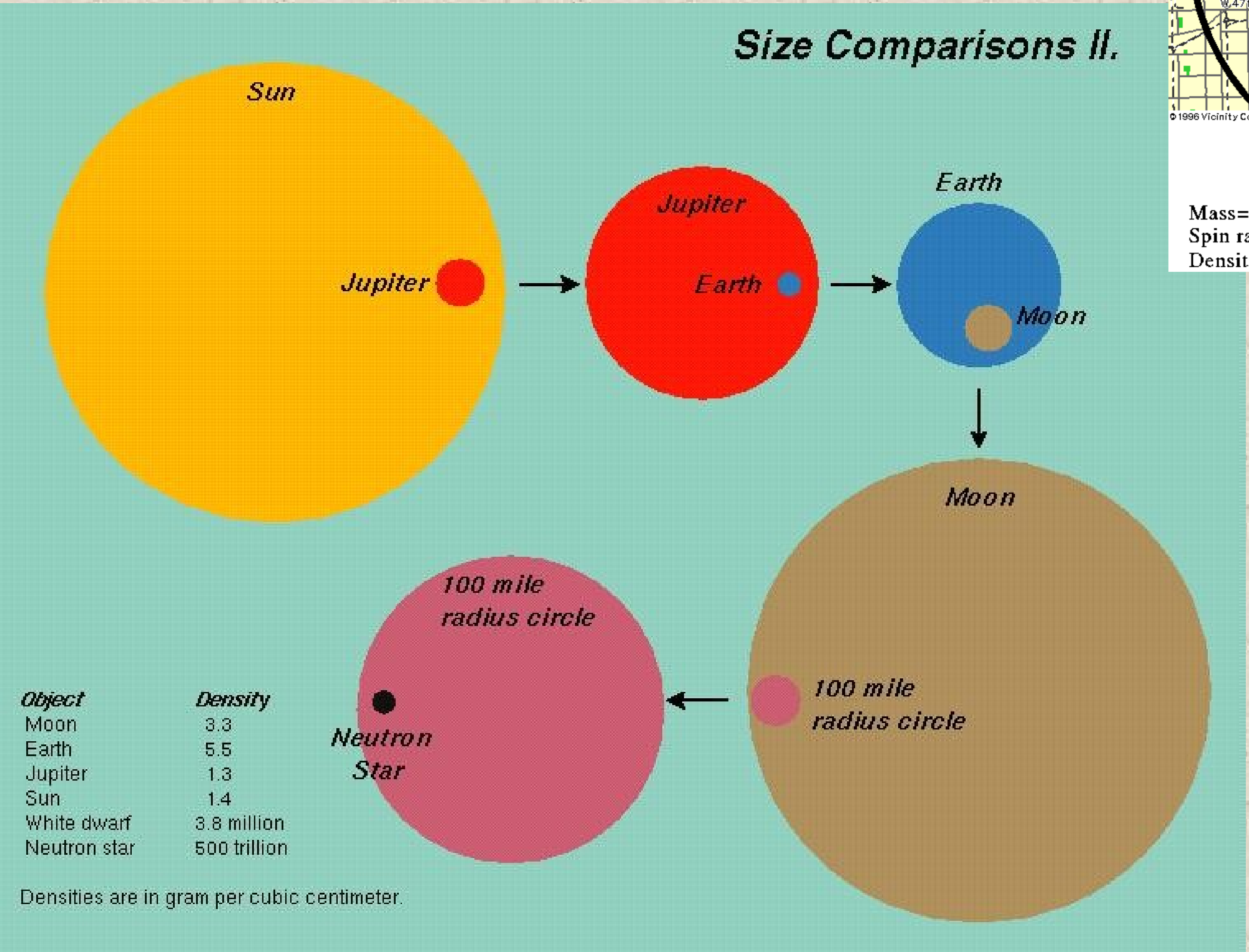
A ***baseball*** has a mass between 5 and 5.25 ounce. Dropping a baseball from space onto a neutron star releases energy equivalent to about ***30 atomic bombs*** the size of the one dropped on Hiroshima.

Neutron Star Magnetic and Electric Fields

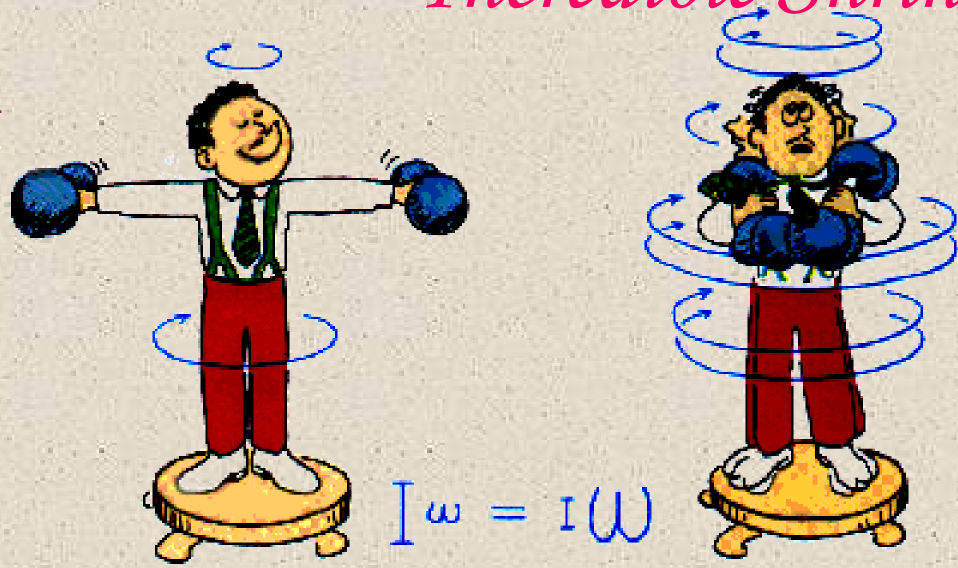
Magnetic fields: Some neutron stars are known to have surface magnetic fields ***trillions*** of times stronger than that at the surface of the Earth, and ***billions*** of times stronger than the magnetic fields concentrated in ***sunspots***.

Electric fields: A rotating magnetized conductor generates an electric field whose strength depends on magnetic field strength and rotation rate. If the neutron star exterior remained a vacuum, the electric field near the surface of the Crab Pulsar would be ***~2 trillion Volts per centimeter***. Some portion of this field is shorted out by the presence of non-neutral plasma, but neutron star magnetospheres remain favored sites for charged particle acceleration and the origin of cosmic rays.

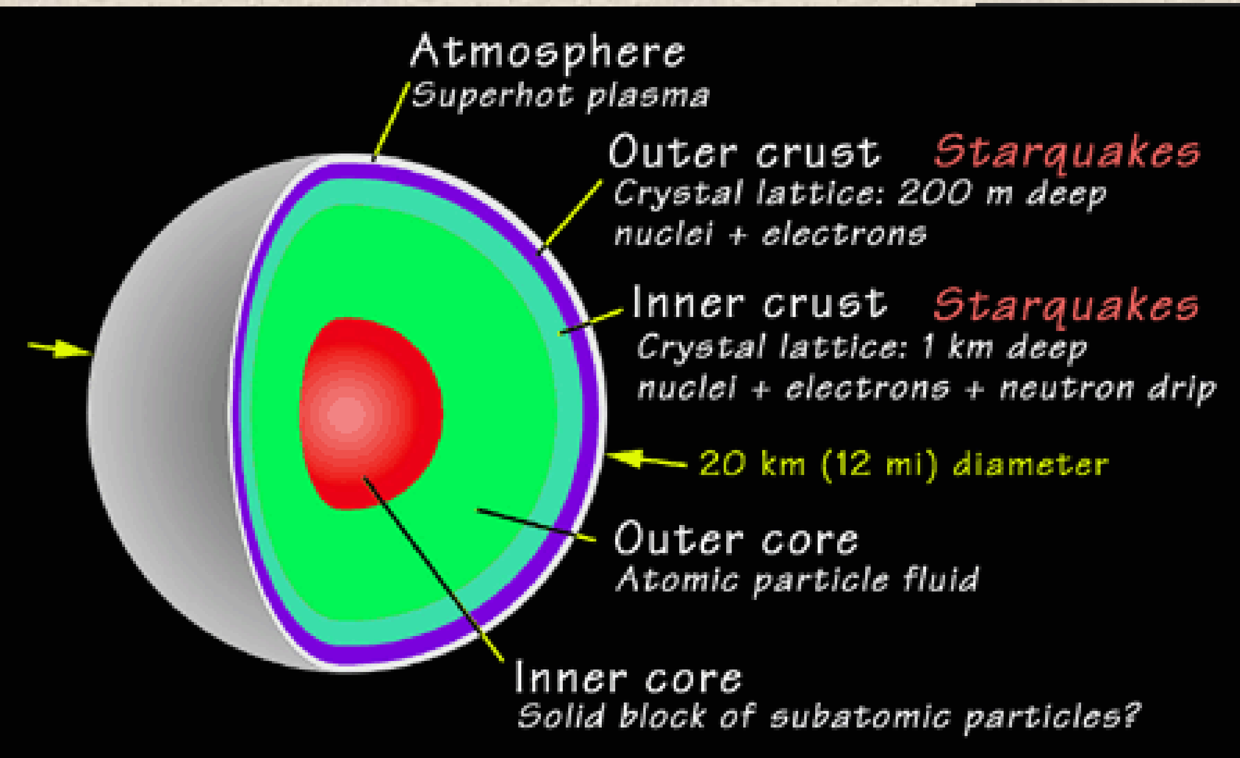
Relative sizes

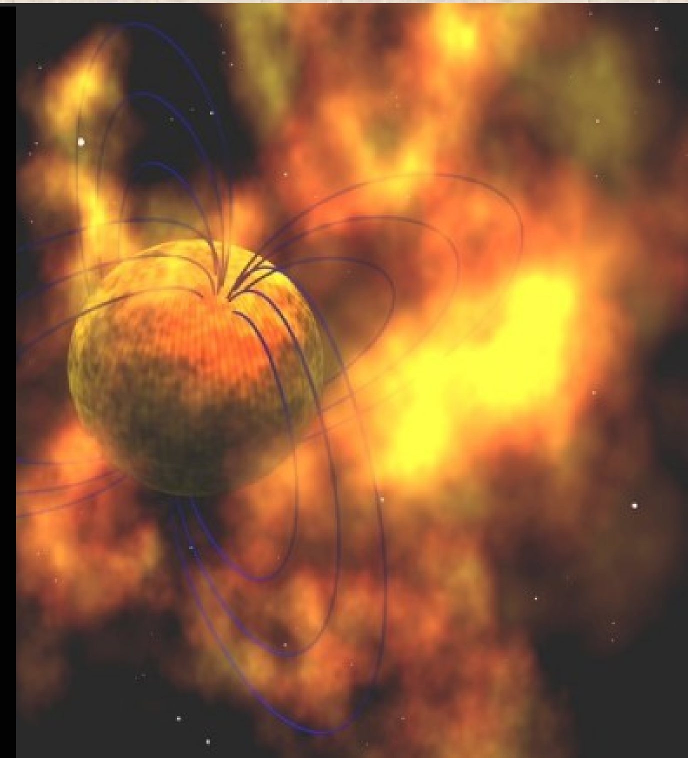


Incredible Shrinking and Growing

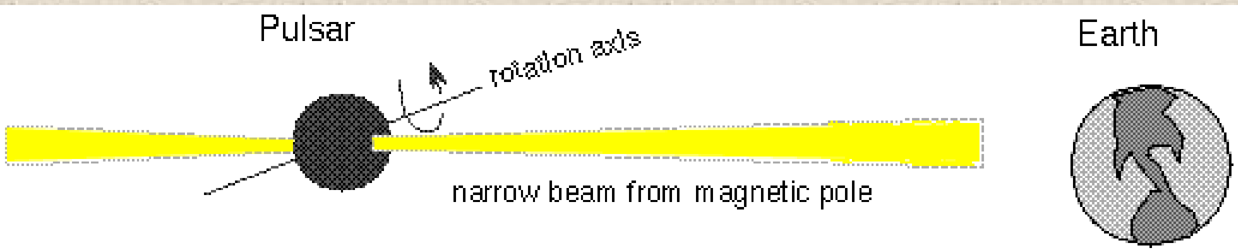


The size of the star is hugely compressed, so the spin is greatly increased and the magnetic field is greatly enhanced. NS are so strong they can spin 1000 times/sec (white dwarfs would fly apart at a few second periods).



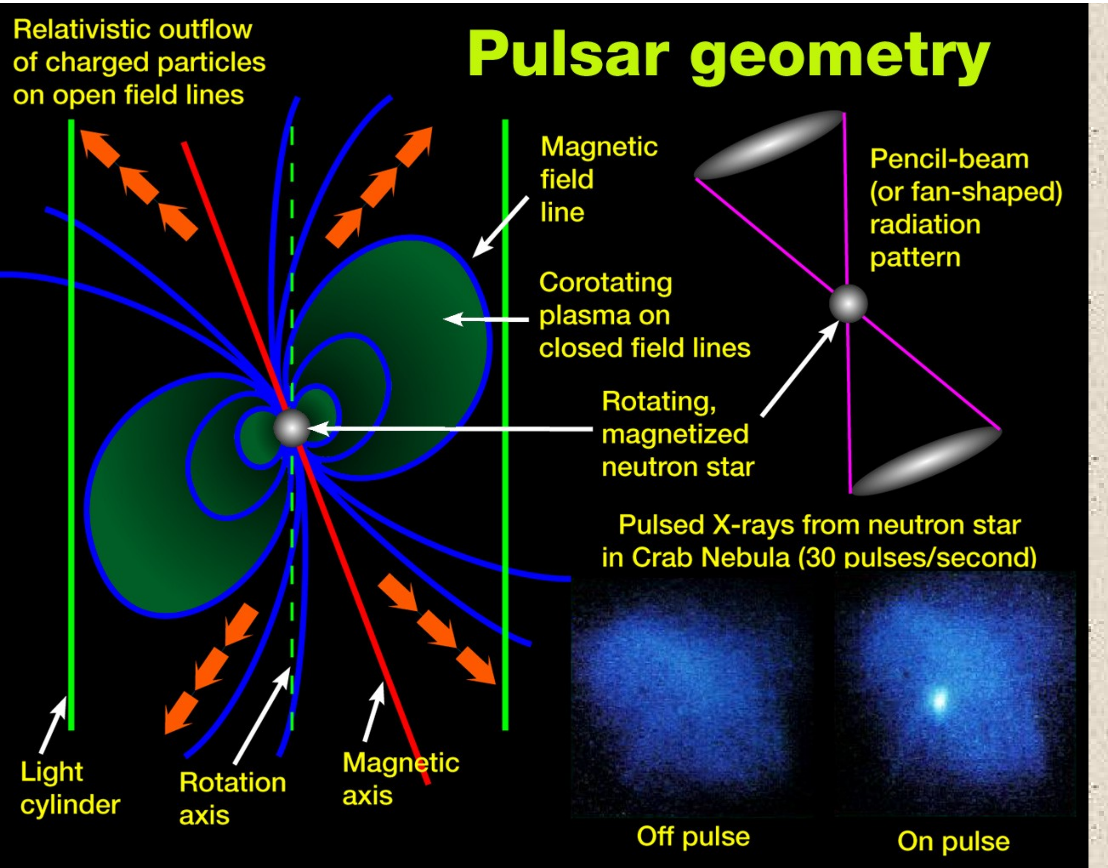


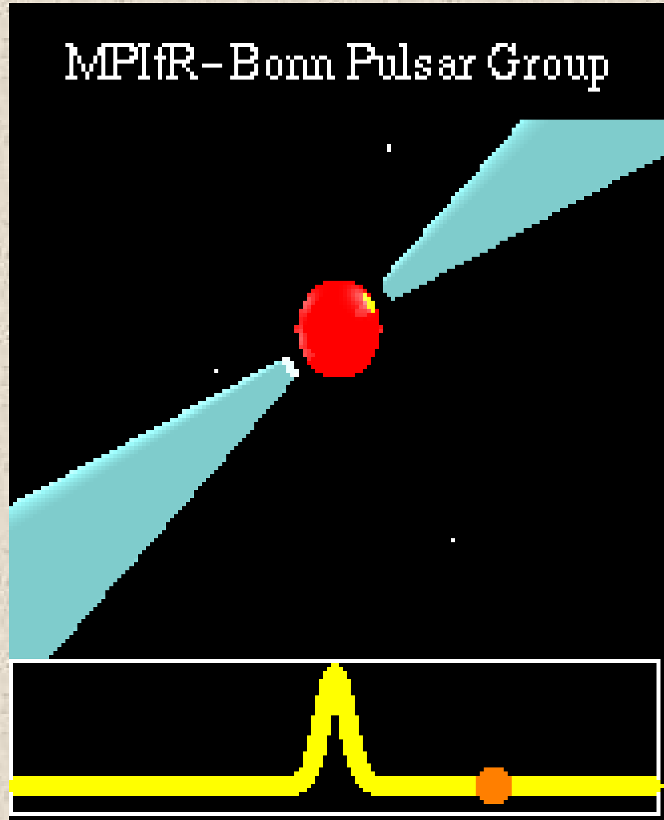
Magnetic poles make a "pulsar"



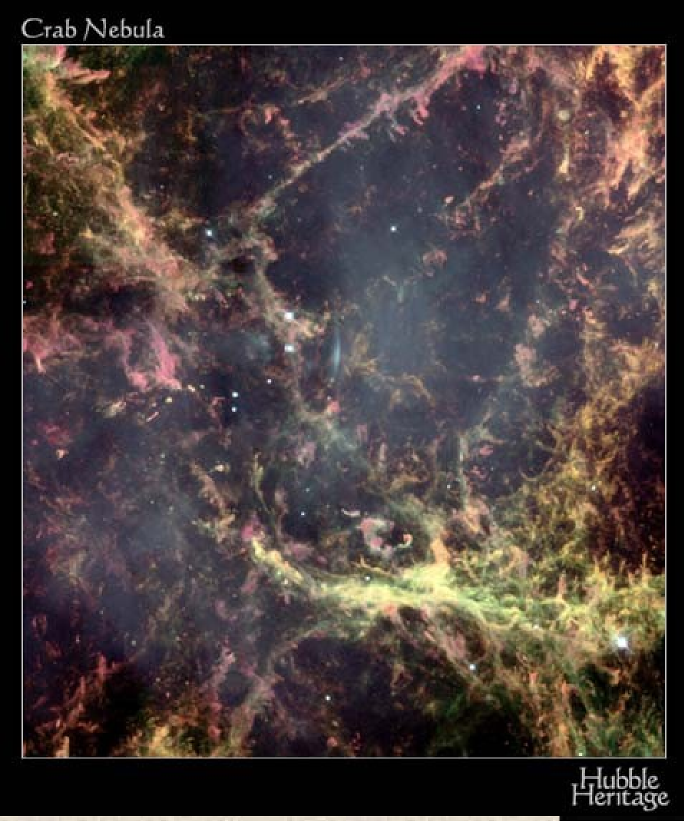
If the narrow synchrotron beam passes over the Earth, we see the neutron star flash on and off like a lighthouse beam does for ships at sea.

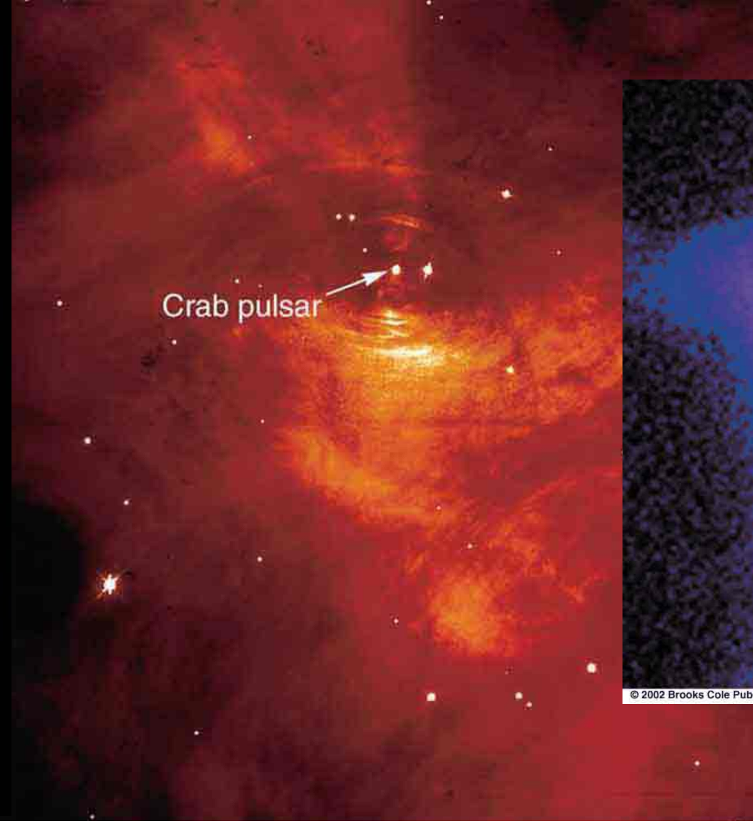
Intense fields with high energy charges flowing along them make a powerful radio beam emitter. This is carried around by the pulsar's rotation, and may pass over the Earth.

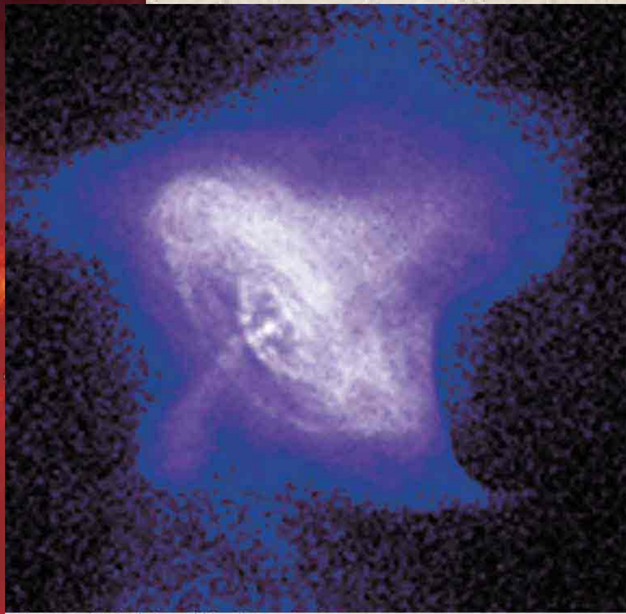


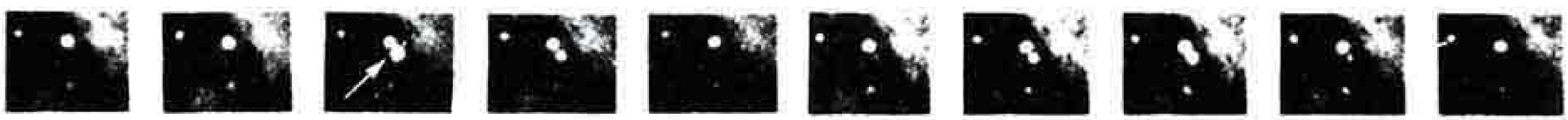


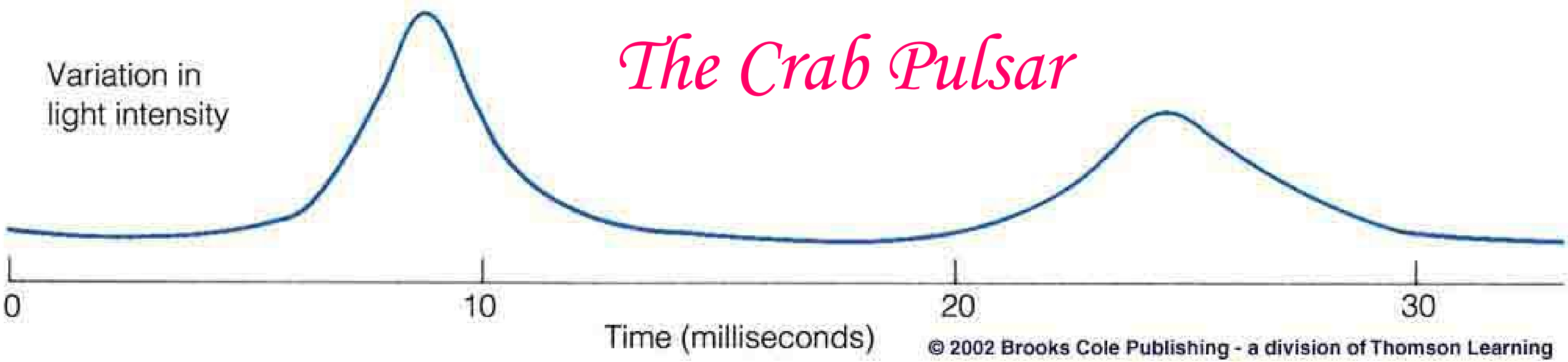
The Crab Pulsar



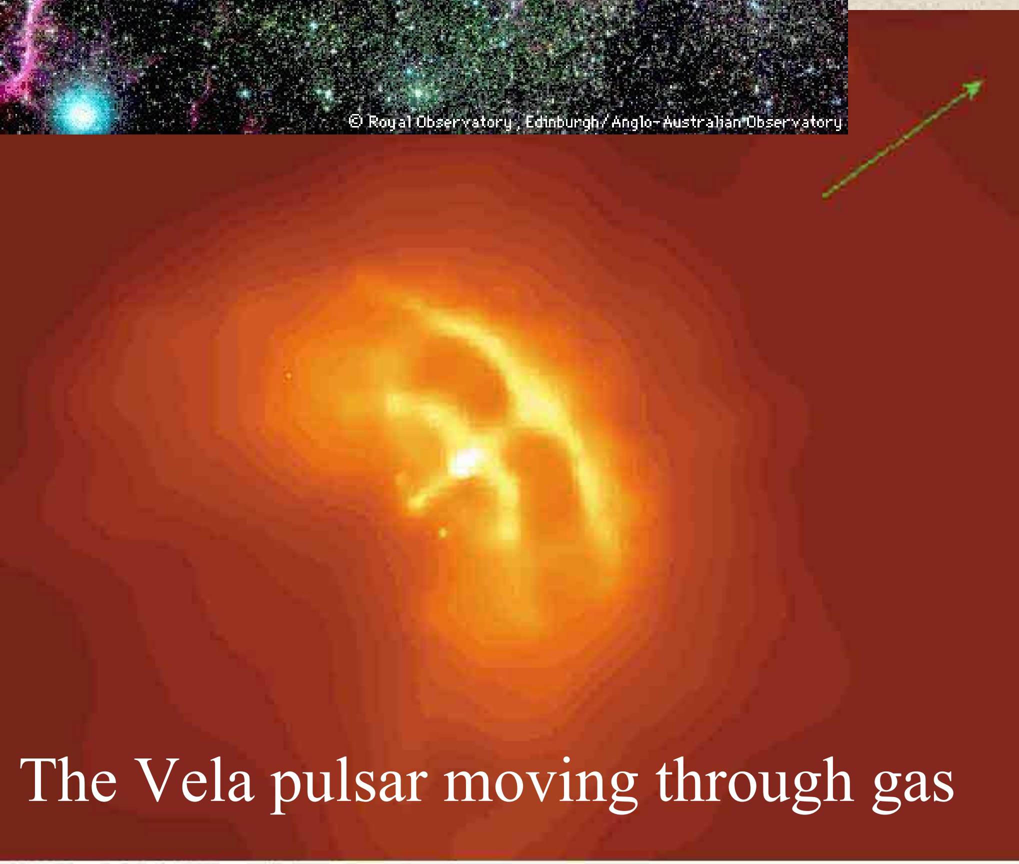






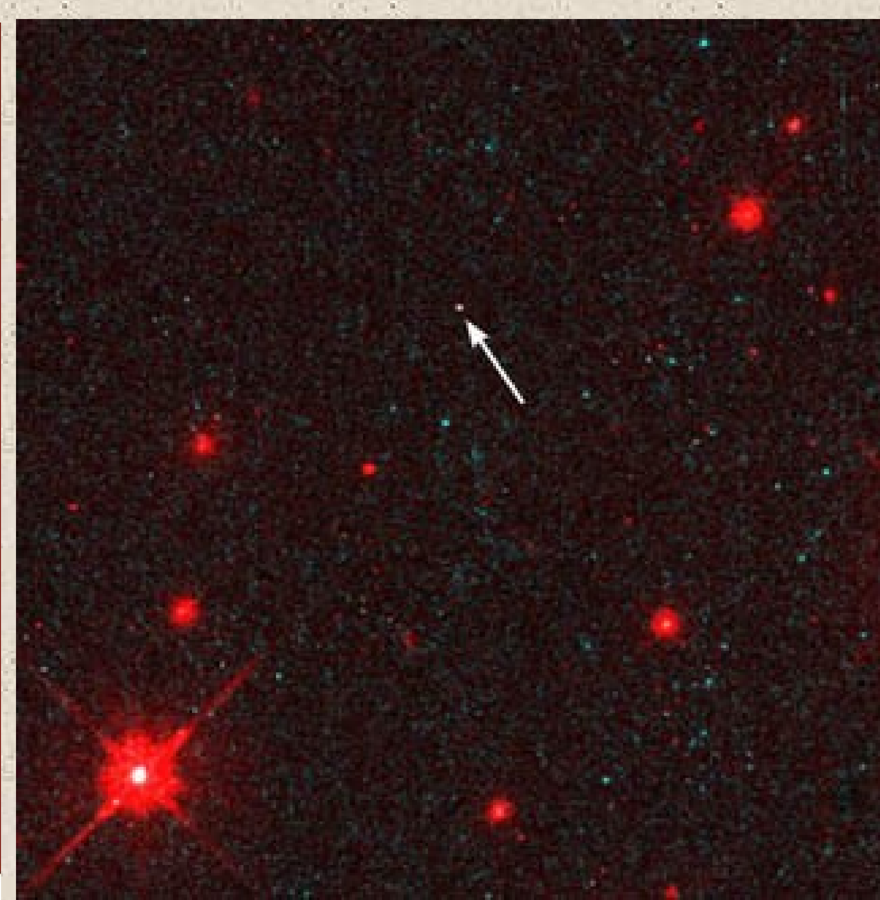


Other Neutron Stars

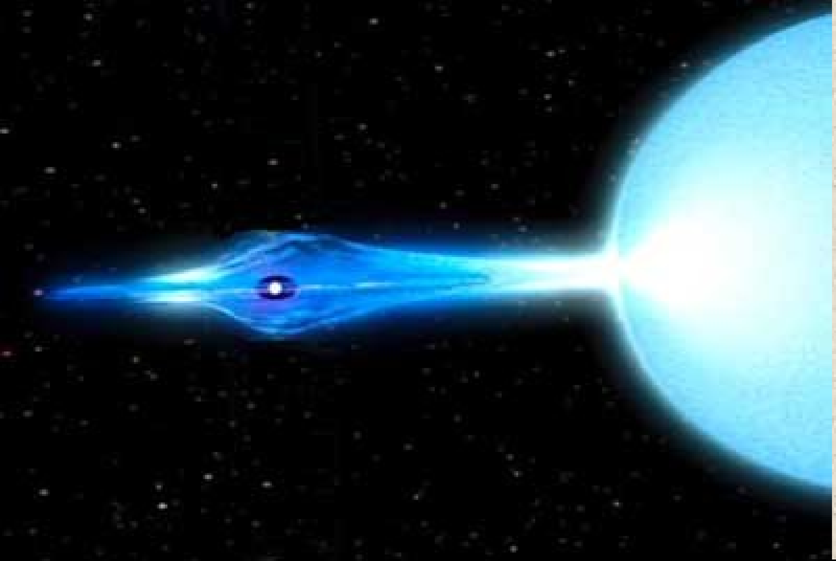


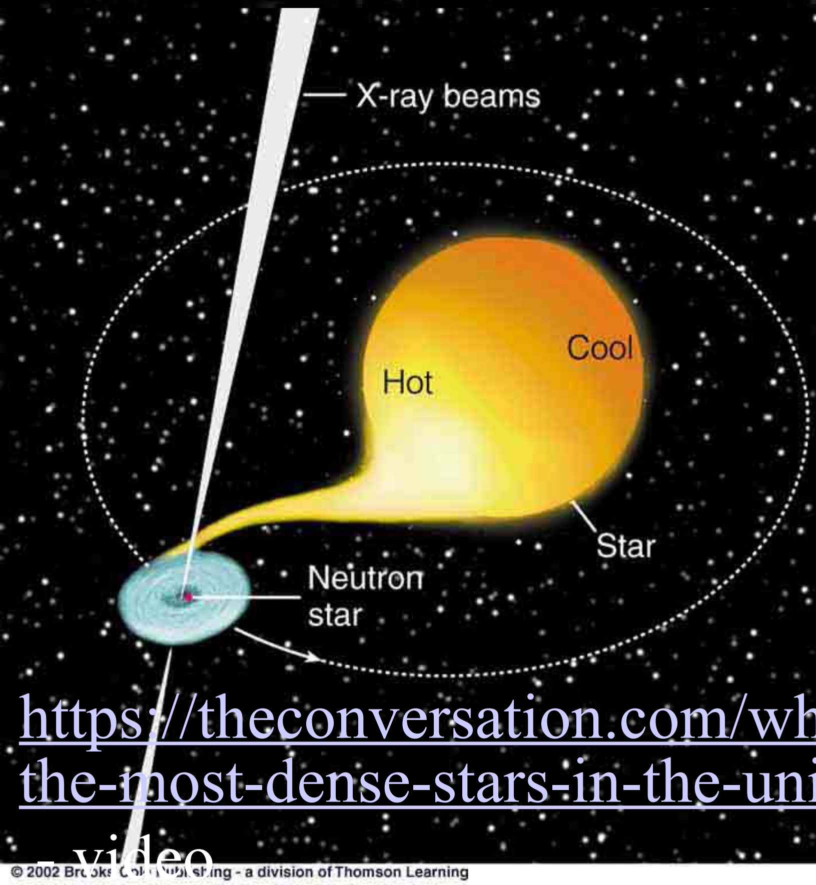
The Vela pulsar moving through gas

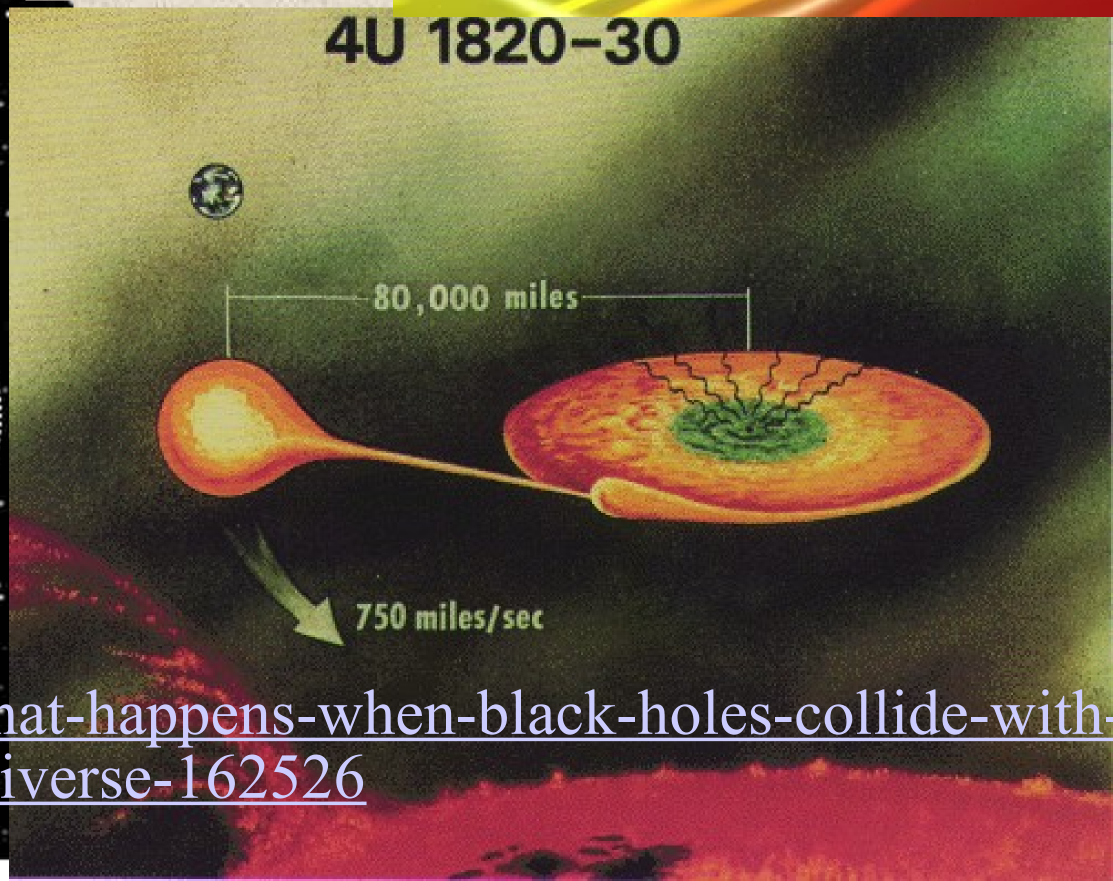
An actual HST image of a nearby neutron star (<100 ly). Surface temperature: 600,000K



Accreting Neutron Stars







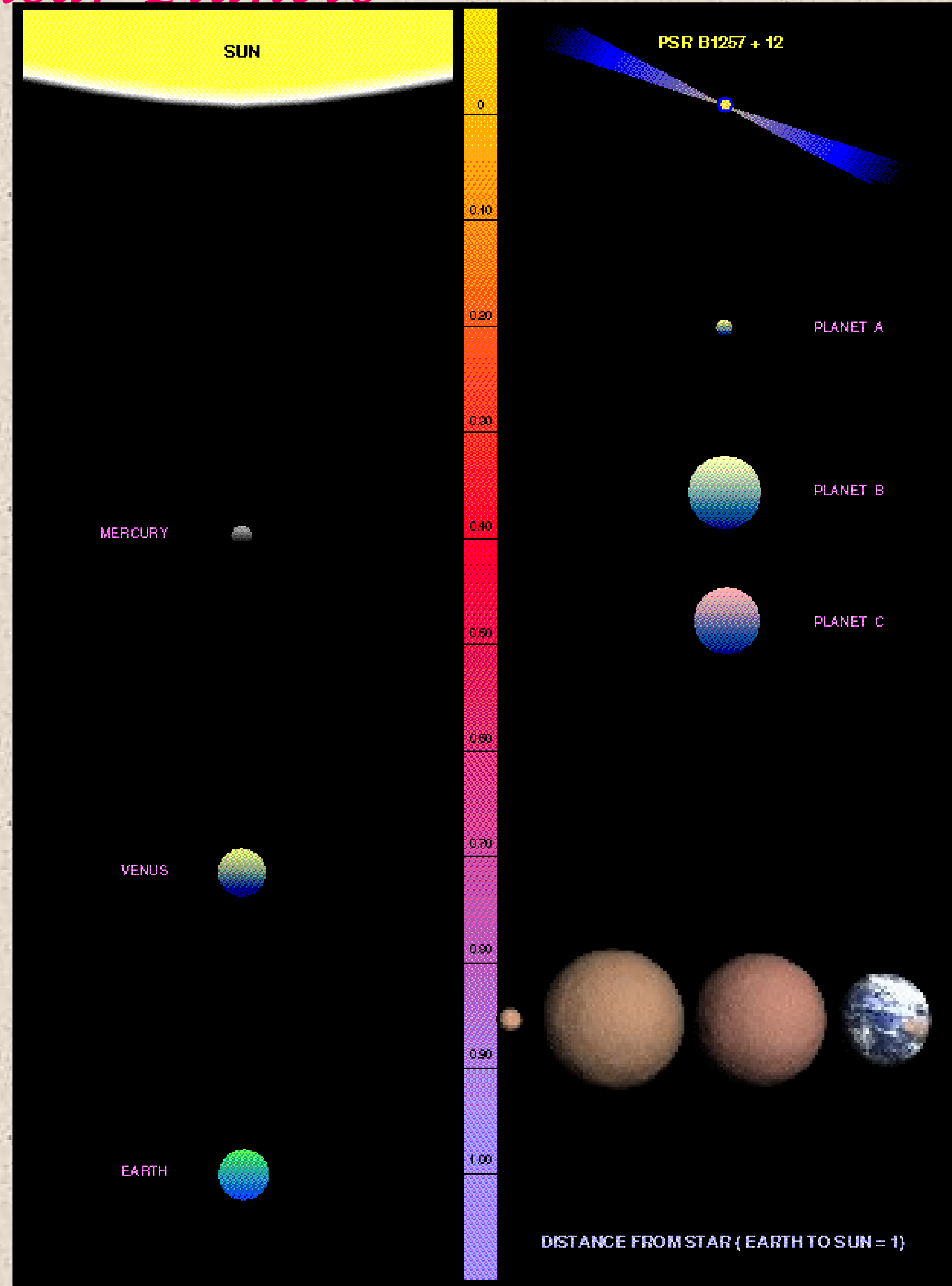
https://theconversation.com/what-happens-when-black-holes-collide-with-the-most-dense-stars-in-the-universe-162526

- video

The Pulsar Planets

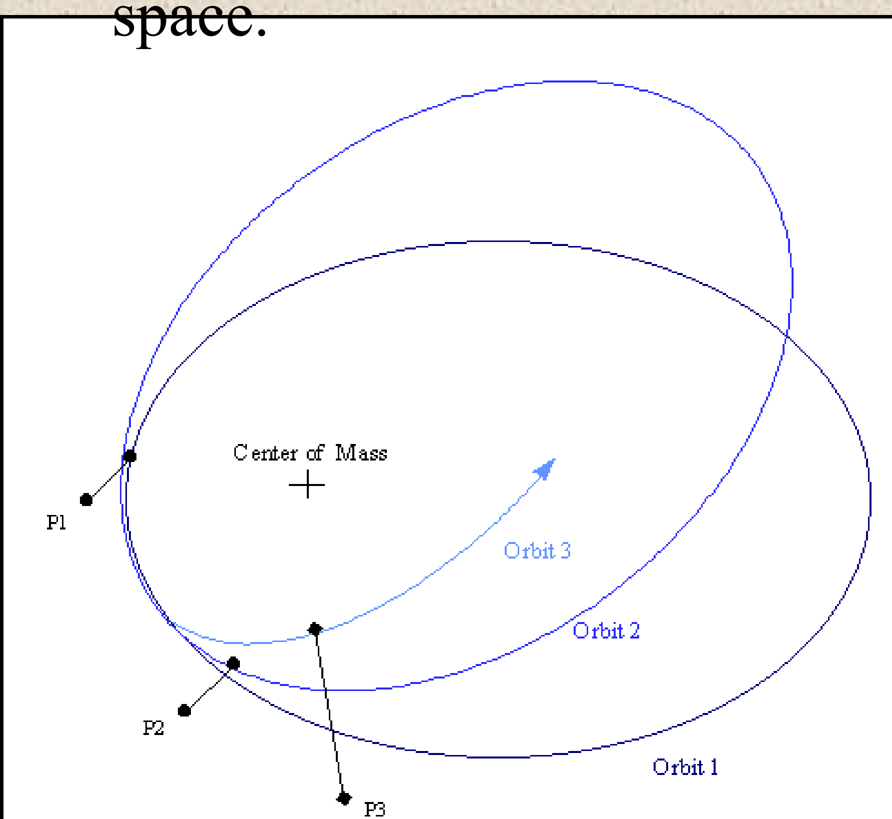
Because of the exquisitely accurate timing signal provided by pulsars, the first extrasolar planets were discovered around a pulsar, and are the only terrestrial extrasolar planets known.

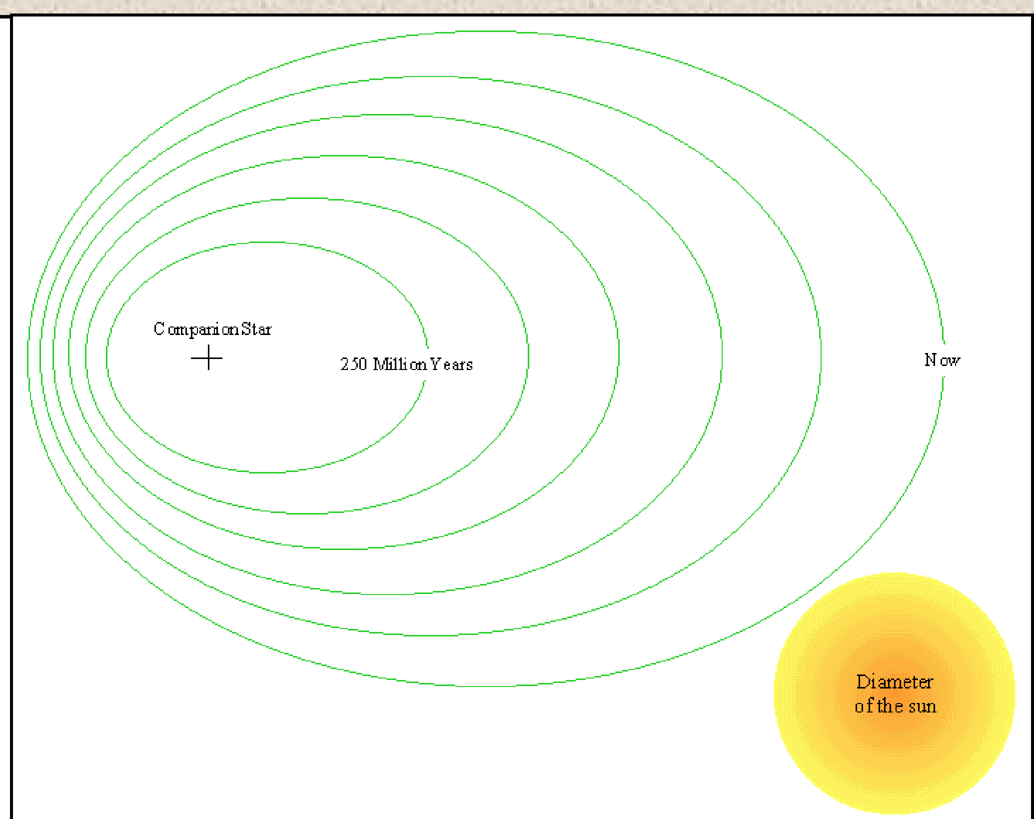
Of course, they aren't exactly Earth-like, given that they must have been formed after the supernova explosion, and their "sun" is a tiny dim object that emits extremely hard radiation and energetic particles…



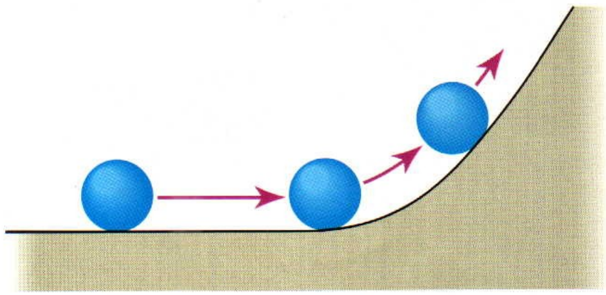
The Binary Pulsar

A system with 2 neutron stars in orbit provides a test of Einstein’s General Relativity (theory of gravitation) [and a Nobel prize for an astronomer (Joe Taylor) and his grad student (Russell Hulse)] It confirms that gravitational radiation is causing the orbit to decay, and measures the orbital precession due to curved space.

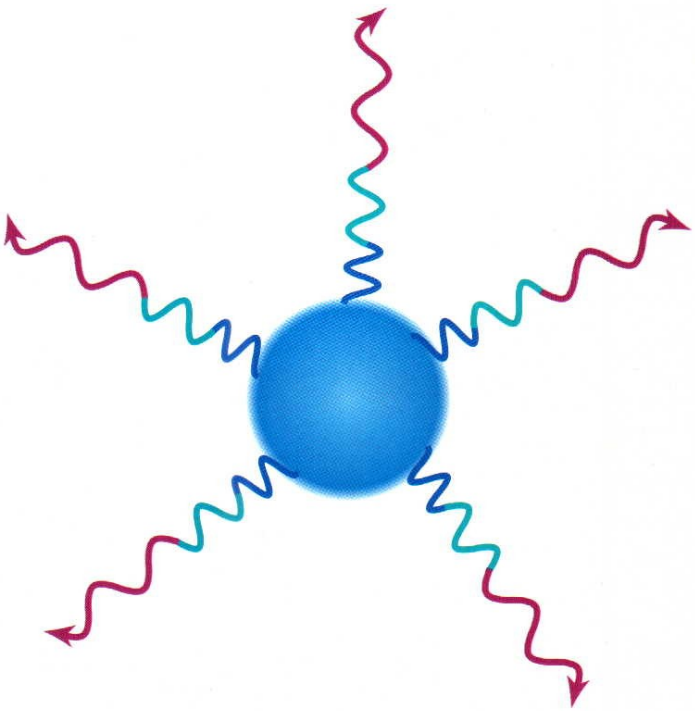




Gravity and Light

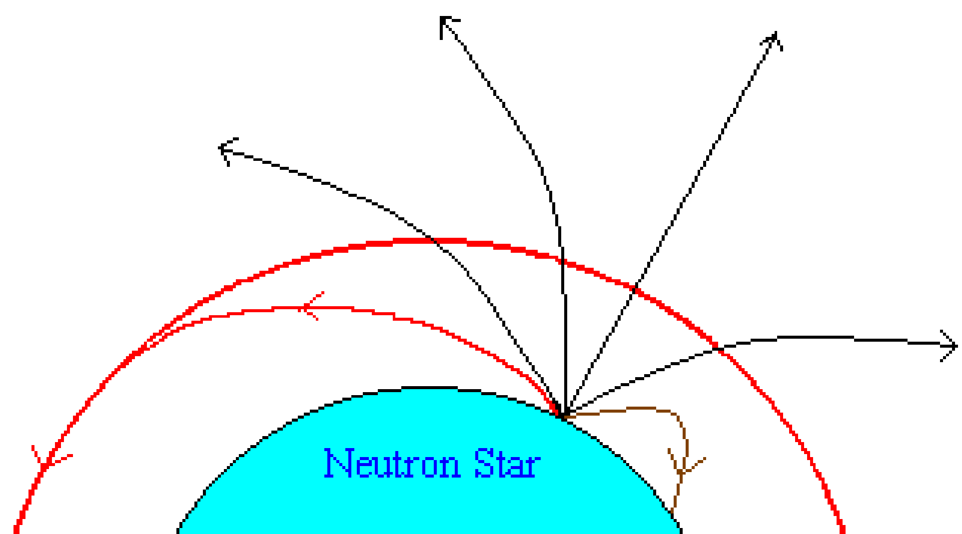


Ball rolling up a hill loses energy and slows down.

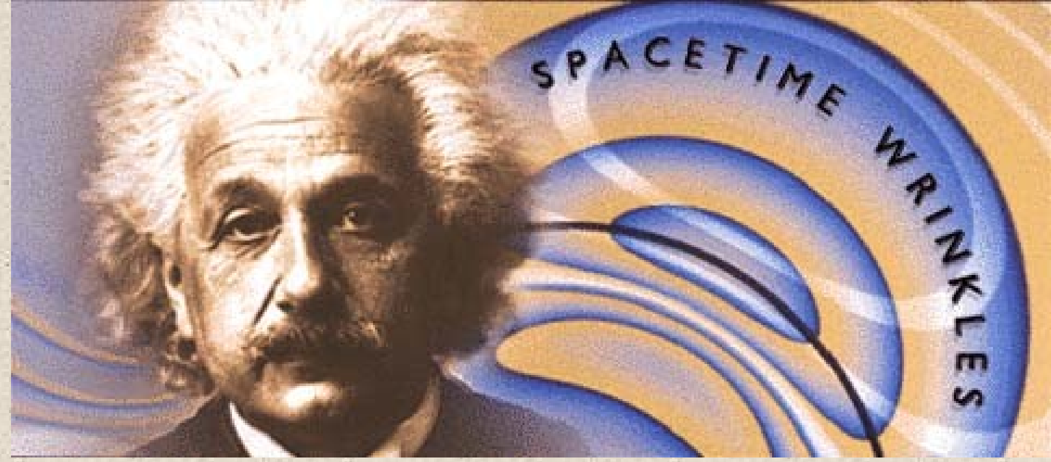


Light moving away from a star loses energy.

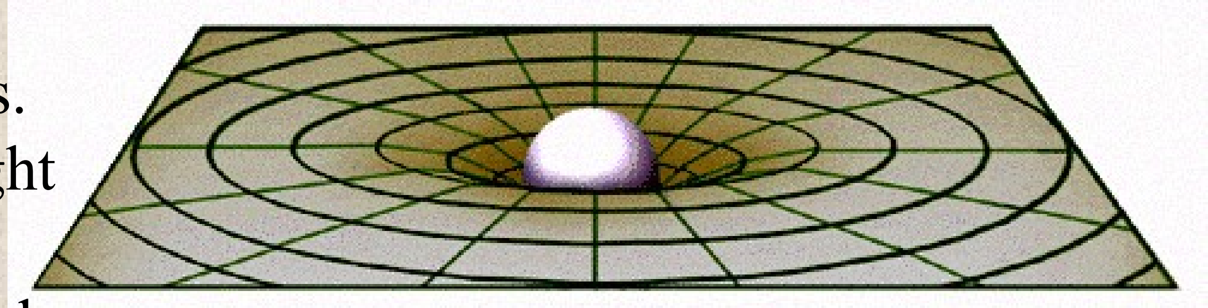
The gravity at a neutron star is extremely strong. One effect is that light leaving the surface is very redshifted (it loses energy). Another is that light rays leaving at an angle follow rather curved paths. Einstein has a novel explanation for this…

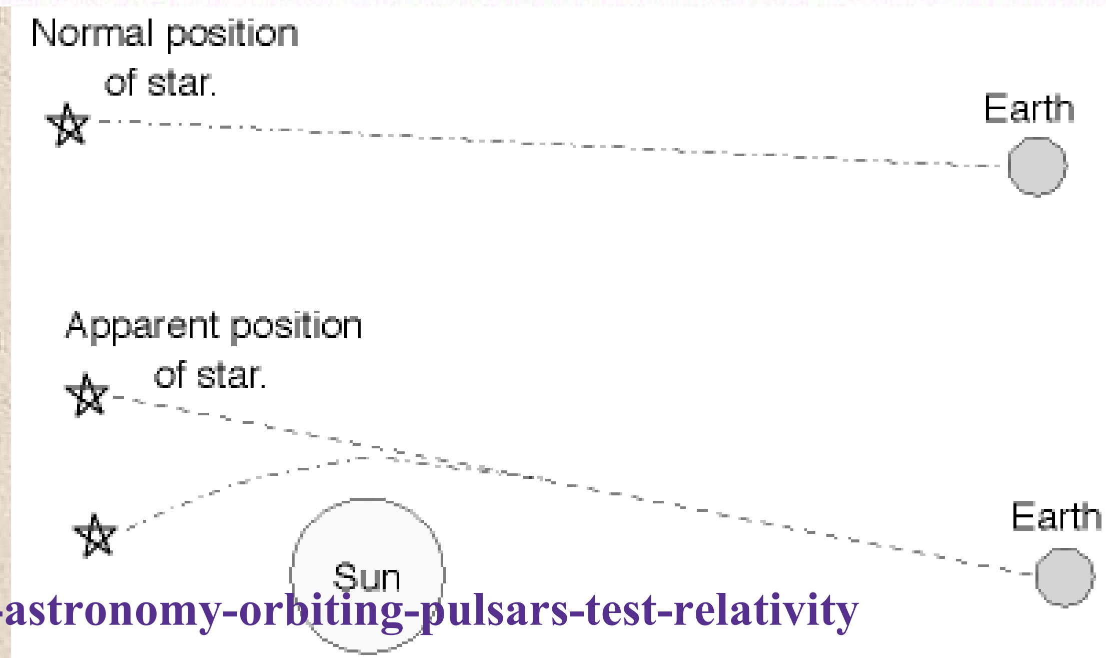


Gravity as a curvature of spacetime



Einstein didn't think of gravity as a force between objects, but as a curving of "straight lines" due to mass. Light always follows straight lines, but these may look curved near masses. Time also slows down near masses (space and time are different parts of "spacetime", which is what gets bent). Both the curving of light and slowing of time are experimentally verified.



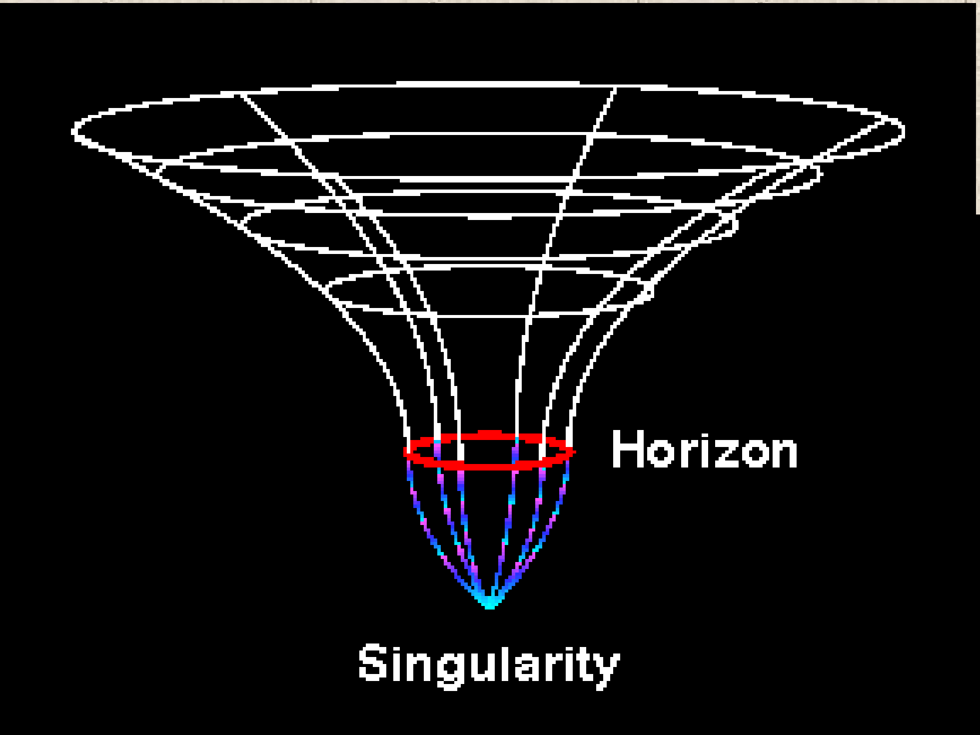


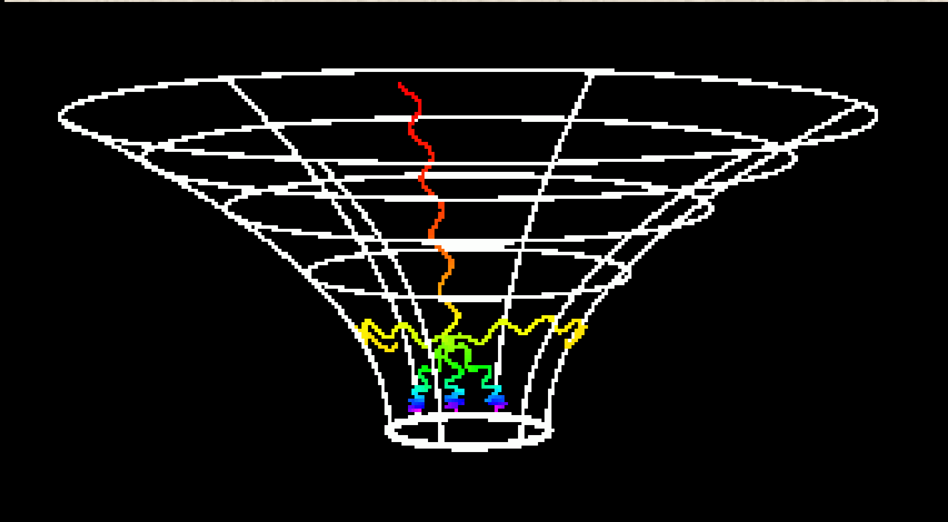
https://www.syfy.com/syfy-wire/bad-astronomy-orbiting-pulsars-test-relativity

Extreme Gravity leads to "Black Holes"

As the object becomes too dense, the "straight lines" of light begin to all be closed onto themselves. The spacetime around the hole gets closed off from the rest of the Universe.

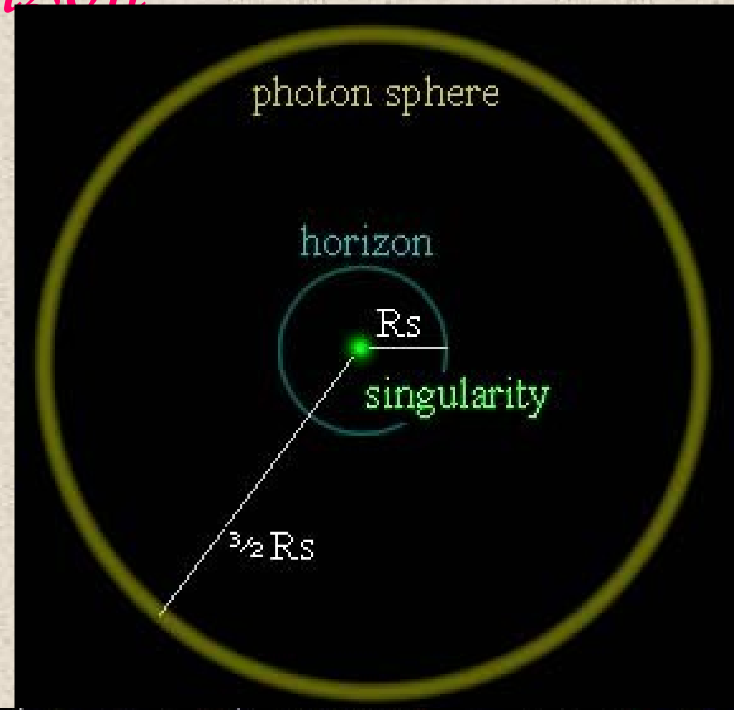
The "exit cone" can pinch off.



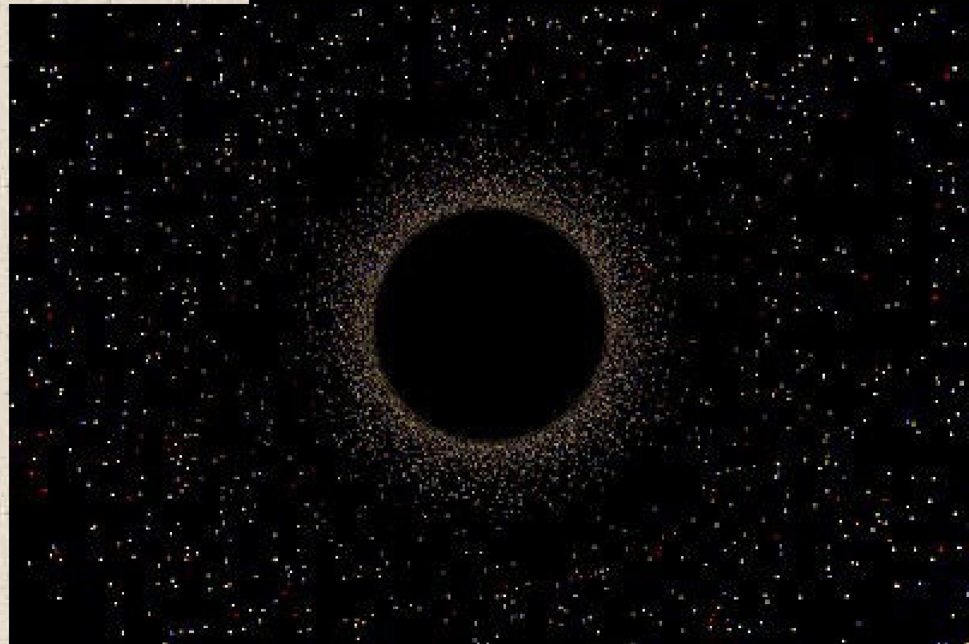


The Event Horizon

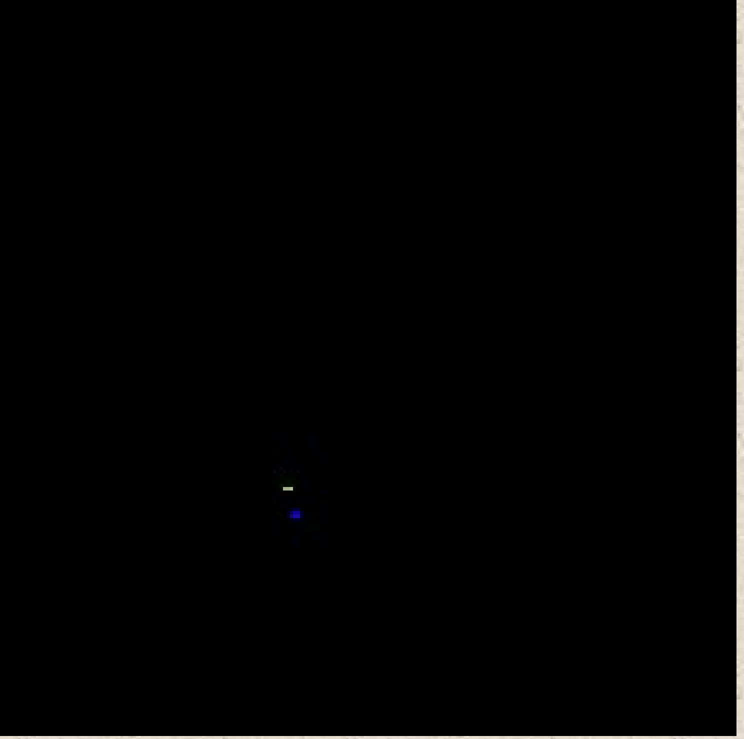
If you calculate the size of an object whose escape velocity is the speed of light, you get the "Schwarzschild radius", which defines the "event horizon". This is the formal size of a black hole (even though there is nothing at that location). It is given by R_s=3km(M_*/M_{sun}). It is the horizon over which you can see no more events. Outside that at 1.5 R_s photons would orbit the hole (the photon sphere).

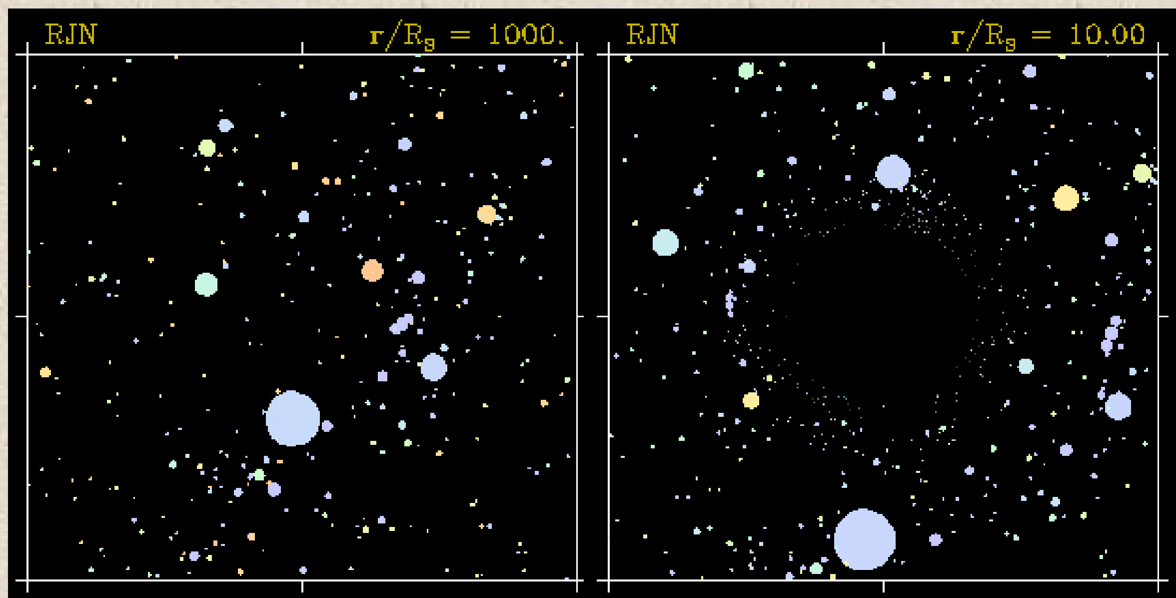


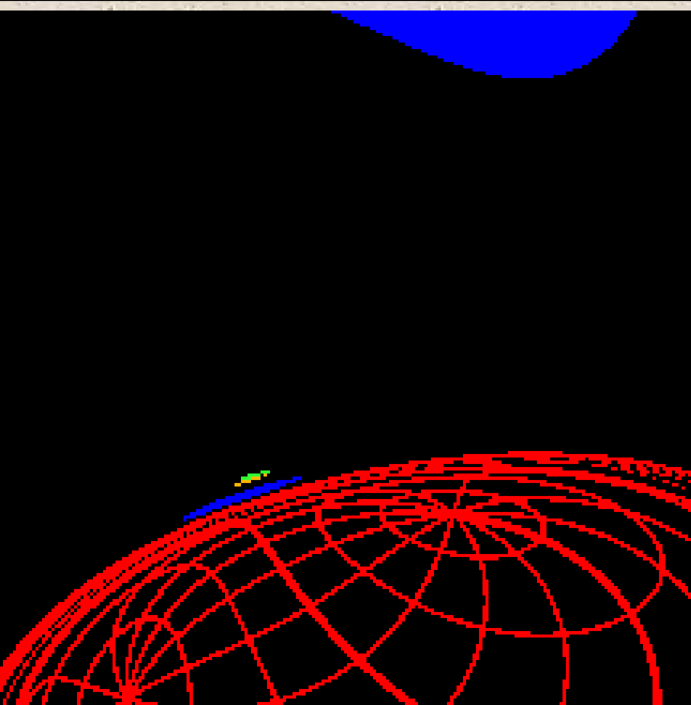
Far from the hole, the gravity is the same as it would be if the star were still there (so no "vacuum cleaner" effect). If the Sun collapsed to a BH, the Earth's orbit would be unaffected. The trick is that you can approach VERY close to the full mass since the object got so dense.



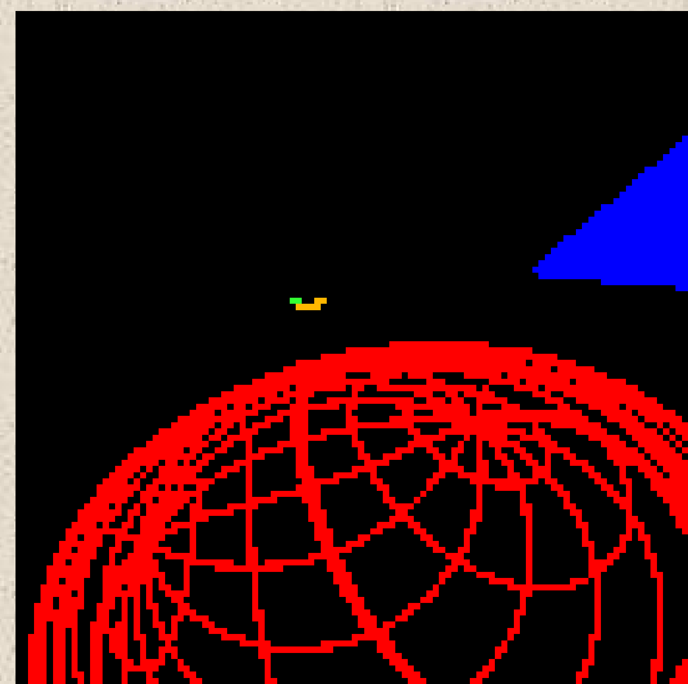
Optical Distortion effects



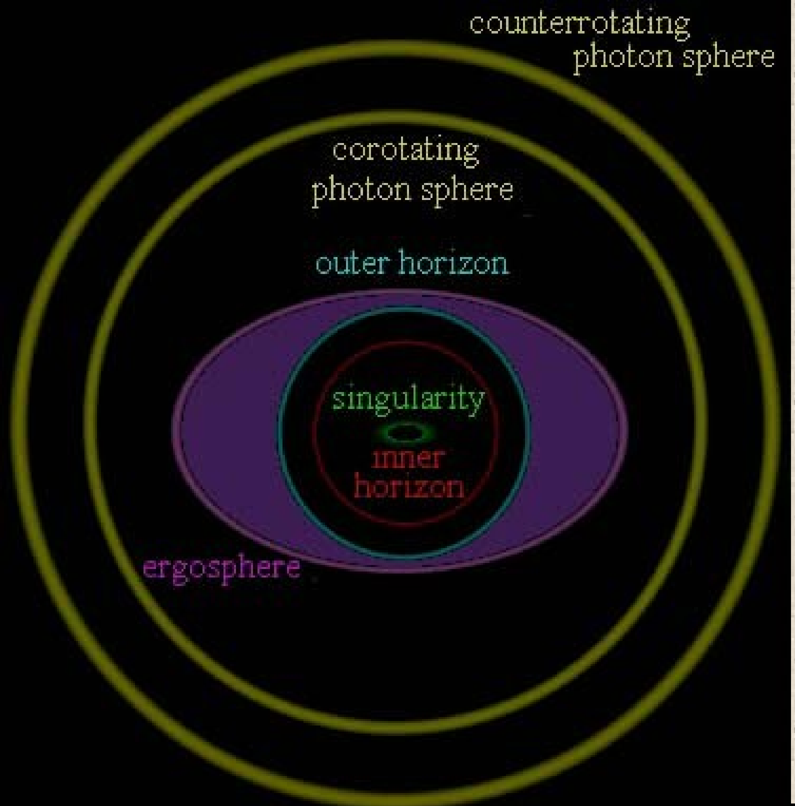




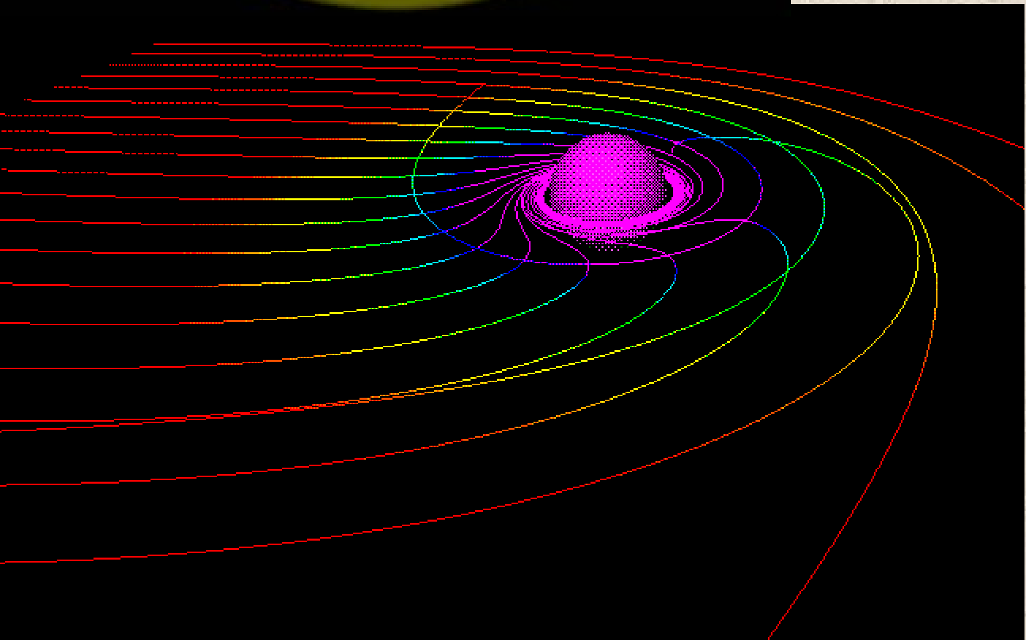
Light bending near the hole can give rise to a very confusing view…

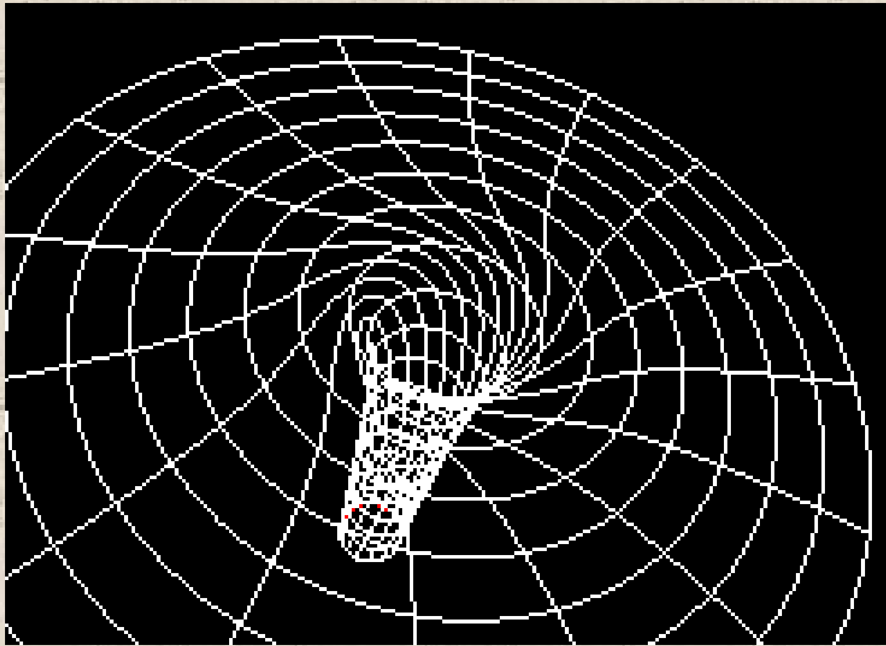


Rotating Black Holes



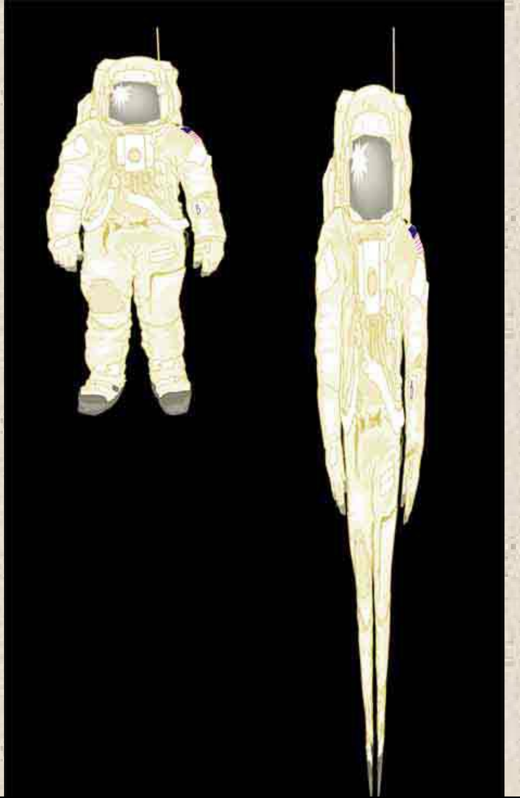
Rotation also leads to “frame-dragging”



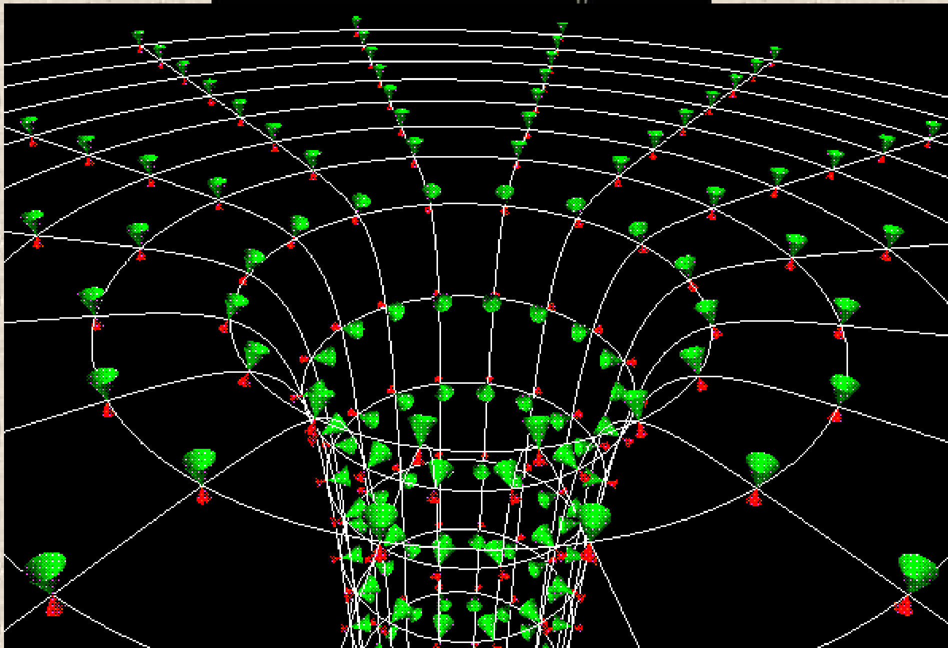


Falling into a Black Hole

Since tidal forces change as the cube of distance, they get extreme near the hole. You are stretched because your feet are nearer than your head, and you are compressed because your shoulders are converging on the singularity. An outside observer sees you getting dimmer and redder, and your watch looks like it runs more slowly. Finally your fading image freezes just outside the event horizon.

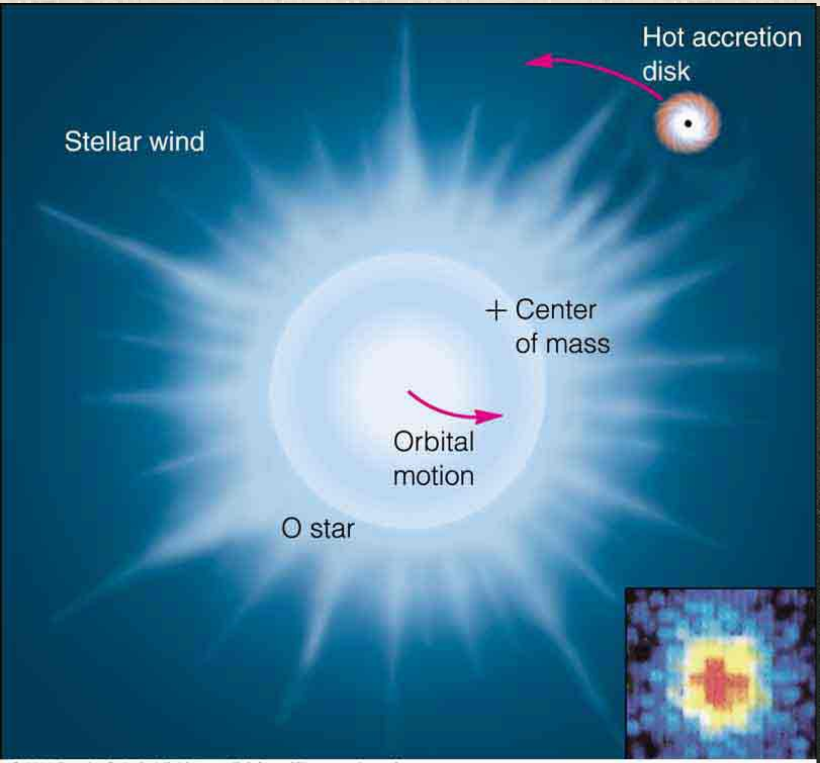


You, on the other hand, seem to fall into the hole in a few more milliseconds. The outside Universe gets pinched into a smaller and smaller angle above your head, and time appears to speed up in it. Of course, you don't actually survive to see this…



Cygnus X-1 : A Black Hole System

As with novae, the presence of a companion which fills its Roche lobe can give rise to an accretion disk around the black hole. As the matter spirals down to the hole, it can become very hot, and emit X-rays which give away the presence of the hole. This is our best means of detecting black holes.



Black Hole accretion

Magnetic fields dragged around the hole can cause material to be flung at high speed out the rotational poles. These relativistic jets are another good sign of accretion onto compact objects (NS or BH).